



Known Streaks Issue

We're fixing a streaks issue this week. Your streaks are safe. Thanks for your patience!

Protected Archives

Besides standalone files, there is also another format of files that can contain not only data, such as an Office document or a PDF, but also other files within them. This format is called an **archive** or **compressed file** that can be protected with a password if necessary.

Let us assume an employee's role in an administrative company and imagine that our customer wants to summarize analysis in different formats, such as Excel, PDF, Word, and a corresponding presentation. One solution would be to send these files individually, but if we extend this example to a large company dealing with several projects running simultaneously, this type of file transfer can become cumbersome and lead to individual files being lost. In these cases, employees often rely on archives, which allow them to split all the necessary files in a structured way according to the projects (often in subfolders), summarize them, and pack them into a single file.

There are many types of archive files. Some common file extensions include, but are not limited to:

tar	gz	rar	zip
vmdb/vmx	cpt	truecrypt	bitlocker
kdbx	luks	deb	7z
pkg	rpm	war	gzip

An extensive list of archive types can be found on [FileInfo.com](https://fileinfo.com). However, instead of manually typing them out, we can also query them using a one-liner, filter them out, and save them to a file if needed. At the time of writing, there are **337** archive file types listed on fileinfo.com.

Download All File Extensions

Protected Archives

```
dbcyp0n@htb[/htb]$ curl -s https://fileinfo.com/filetypes/compressed | html2text | awk '{print tolower($1)}' | gr
.mint
.htmi
.tpsr
.mpkg
.arduboy
.ice
.sifz
.fzpz
.rar
.comppkg.hauptwerk.rar
...SNIP...
```

It is important to note that not all of the above archives support password protection. Other tools are often used to protect the corresponding archives with a password. For example, with **tar**, the tool **openssl** or **gpg** is used to encrypt the archives.

Cracking Archives

Given the number of different archives and the combination of tools, we will show only some of the possible ways to crack specific

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The .zip format is often heavily used in Windows environments to compress many files into one file. The procedure we have already seen remains the same except for using a different script to extract the hashes.

Cracking ZIP

Using zip2john

Protected Archives
<pre>dbcypthon@htb[/htb]\$ zip2john ZIP.zip > zip.hash</pre>
<pre>ver 2.0 efh 5455 efh 7875 ZIP.zip/flag.txt PKZIP Encr: 2b chk, TS_chk, cmplen=42, decmplen=30, crc=490E7510</pre>

By extracting the hashes, we will also see which files are in the ZIP archive.

Viewing the Contents of zip.hash

Protected Archives
<pre>dbcypthon@htb[/htb]\$ cat zip.hash</pre>
<pre>ZIP.zip/customers.csv:\$pkzip2\$1*2*2*0*2a*1e*490e7510*0*42*0*2a*490e*409b*ef1e7feb7c1cf701a6ada7132e6a5c6c84c032401</pre>

Once we have extracted the hash, we can now use **john** again to crack it with the desired password list. Because if **john** cracks it successfully, it will show us the corresponding password that we can use to open the ZIP archive.

Cracking the Hash with John

Protected Archives
<pre>dbcypthon@htb[/htb]\$ john --wordlist=rockyou.txt zip.hash</pre>
<pre>Using default input encoding: UTF-8 Loaded 1 password hash (PKZIP [32/64]) Will run 2 OpenMP threads Press 'q' or Ctrl-C to abort, almost any other key for status 1234 (ZIP.zip/customers.csv) 1g 0:00:00:00 DONE (2022-02-09 09:18) 100.0g/s 250600p/s 250600c/s 250600C/s 123456..1478963 Use the "--show" option to display all of the cracked passwords reliably Session completed</pre>

Viewing the Cracked Hash

Protected Archives
<pre>dbcypthon@htb[/htb]\$ john zip.hash --show</pre>
<pre>ZIP.zip/customers.csv:1234:customers.csv:ZIP.zip::ZIP.zip</pre>
<pre>1 password hash cracked, 0 left</pre>

Cracking OpenSSL Encrypted Archives

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Listing the Files

Protected Archives
<pre>dbcyph0n@htb[/htb]\$ ls</pre>
<pre>GZIP.zip rockyou.txt</pre>

Using file

Protected Archives
<pre>dbcyph0n@htb[/htb]\$ file GZIP.zip</pre>
<pre>GZIP.zip: openssl enc'd data with salted password</pre>

When cracking OpenSSL encrypted files and archives, we can encounter many different difficulties that will bring many false positives or even fail to guess the correct password. Therefore, the safest choice for success is to use the `openssl` tool in a `for-loop` that tries to extract the files from the archive directly if the password is guessed correctly.

The following one-liner will show many errors related to the GZIP format, which we can ignore. If we have used the correct password list, as in this example, we will see that we have successfully extracted another file from the archive.

Using a for-loop to Display Extracted Contents

Protected Archives
<pre>dbcyph0n@htb[/htb]\$ for i in \$(cat rockyou.txt);do openssl enc -aes-256-cbc -d -in GZIP.zip -k \$i 2>/dev/null tar</pre>
<pre>gzip: stdin: not in gzip format tar: Child returned status 1 tar: Error is not recoverable: exiting now</pre>
<pre>gzip: stdin: not in gzip format tar: Child returned status 1 tar: Error is not recoverable: exiting now</pre>
<pre><SNIP></pre>

Once the for-loop has finished, we can look in the current folder again to check if the cracking of the archive was successful.

Listing the Contents of the Cracked Archive

Protected Archives
<pre>dbcyph0n@htb[/htb]\$ ls</pre>
<pre>customers.csv GZIP.zip rockyou.txt</pre>

Cracking BitLocker Encrypted Drives

[BitLocker](#) is an encryption program for entire partitions and external drives. Microsoft developed it for the Windows operating system. It



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Virtual drives are often created in which personal information, notes, and documents are stored on the computer or laptop provided by the company to prevent access to this information by third parties. Again, we can use a script called `bitlocker2john` to extract the hash we need to crack. [Four different hashes](#) will be extracted, which can be used with different Hashcat hash modes. For our example, we will work with the first one, which refers to the BitLocker password.

Using bitlocker2john

Protected Archives

```
dbcyp0n@htb[/htb]$ bitlocker2john -i Backup.vhd > backup.hashes
dbcyp0n@htb[/htb]$ grep "bitlocker\$0" backup.hashes > backup.hash
dbcyp0n@htb[/htb]$ cat backup.hash

$bitlocker$0$16$02b329c0453b9273f2fc1b927443b5fe$1048576$12$00b0a67f961dd80103000000$60$d59f37e...SNIP...70696f7ea
```

Both `John` and `Hashcat` can be used for this purpose. This example will look at the procedure with `Hashcat`. The Hashcat mode for cracking BitLocker hashes is `-m 22100`. So we provide Hashcat with the file with the one hash, specify our password list, and specify the hash mode. Since this is robust encryption (`AES`), cracking can take some time, depending on the hardware used. Additionally, we can specify the filename in which the result should be stored.

Using hashcat to Crack backup.hash

Protected Archives

```
dbcyp0n@htb[/htb]$ hashcat -m 22100 backup.hash /opt/useful/seclists/Passwords/Leaked-Databases/rockyou.txt -o ba
hashcat (v6.1.1) starting...

<SNIP>

$bitlocker$0$16$02b329c0453b9273f2fc1b927443b5fe$1048576$12$00b0a67f961dd80103000000$60$d59f37e70696f7eab6b8f95ae9

Session.....: hashcat
Status.....: Cracked
Hash.Name.....: BitLocker
Hash.Target.....: $bitlocker$0$16$02b329c0453b9273f2fc1b927443b5fe$10...8ec54f
Time.Started.....: Wed Feb 9 11:46:40 2022 (1 min, 42 secs)
Time.Estimated...: Wed Feb 9 11:48:22 2022 (0 secs)
Guess.Base.....: File (/opt/useful/seclists/Passwords/Leaked-Databases/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 28 H/s (8.79ms) @ Accel:32 Loops:4096 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests
Progress.....: 2880/6163 (46.73%)
Rejected.....: 0/2880 (0.00%)
Restore.Point....: 2816/6163 (45.69%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:1044480-1048576
Candidates.#1....: chemical -> secrets

Started: Wed Feb 9 11:46:35 2022
Stopped: Wed Feb 9 11:48:23 2022
```

Viewing the Cracked Hash

Protected Archives

```
dbcyp0n@htb[/htb]$ cat backup.cracked

$bitlocker$0$16$02b329c0453b9273f2fc1b927443b5fe$1048576$12$00b0a67f961dd80103000000$60$d59f37e70696f7eab6b8f95ae9
```

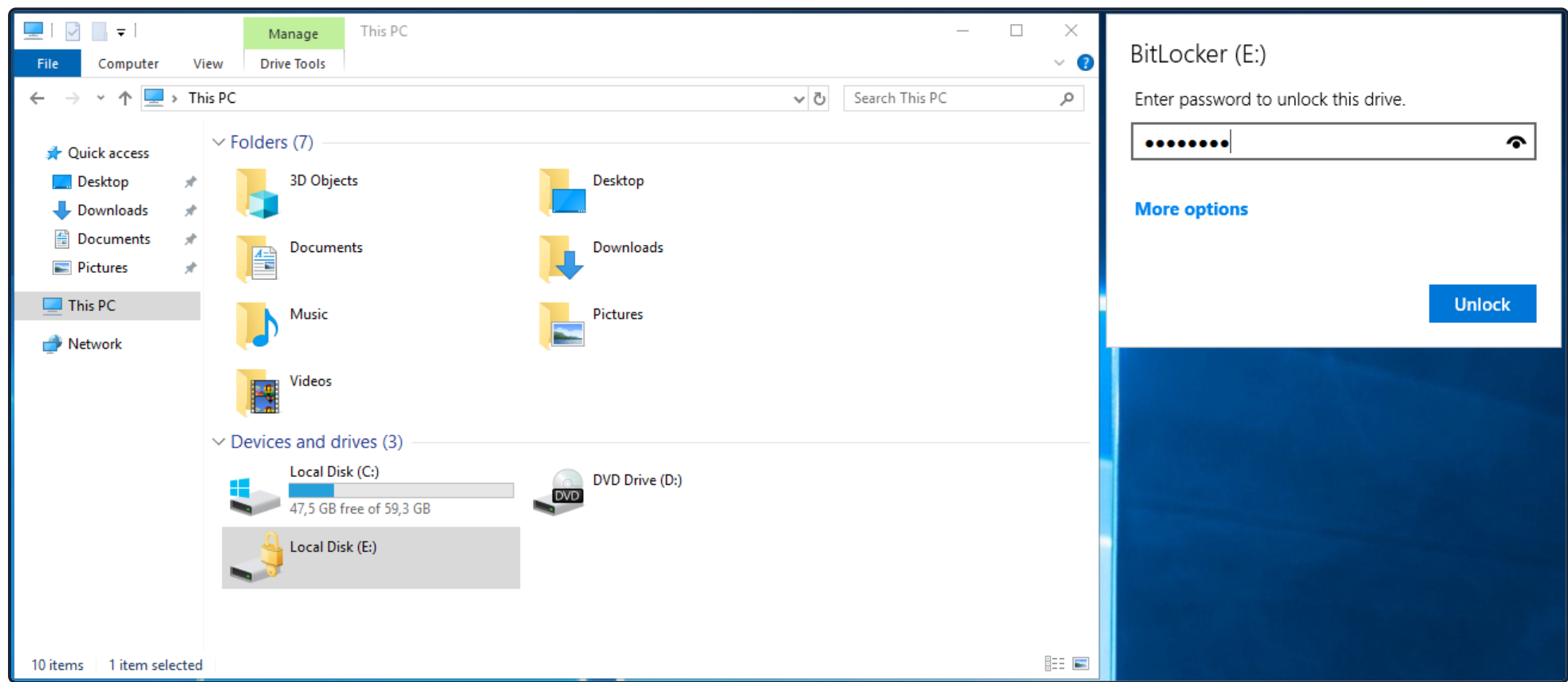


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protected, Windows will show us an error. After mounting, we can again double-click BitLocker to prompt us for the password.

Windows - Mounting BitLocker VHD



VPN Servers

⚠ Warning: Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.

All VM instances associated with the old VPN Server will be terminated when switching to a new VPN server.

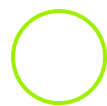
Existing PwnBox instances will automatically switch to the new VPN server.

EU Academy 3Medium Load

PROTOCOL

● UDP 1337 ● TCP 443

DOWNLOAD VPN CONNECTION FILE



Connect to Pwnbox

Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU

146ms

Terminate Pwnbox to switch location



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∞ / 1 spawns left

Waiting to start...

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.218.255 (ACADEMY-PWATTACKS-NIX01)

Life Left: 105 minute(s)

+

Terminate

×

Cheat Sheet

Download VPN Connection File

+ 0

 Use the cracked password of the user Kira, log in to the host, and read the Notes.zip file containing the flag. Then, submit the flag as the answer.

Submit your answer here...

+10 Streak pts

Submit

Show Solution

← Previous

Next →

Cheat Sheet

Resources

Go to Questions

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Theory of Protection	✓
Credential Storage	✓
John The Ripper	✓

Remote Password Attacks

Network Services	✓
Password Mutations	✓
Password Reuse / Default Passwords	✓



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- Attacking LSASS
-

Attacking Active Directory & NTDS.ditCredential Hunting in Windows

Linux Local Password Attacks

- Credential Hunting in Linux
-

Passwd, Shadow & Opasswd

Windows Lateral Movement

- Pass the Hash (PtH)
-

Pass the Ticket (PtT) from WindowsPass the Ticket (PtT) from Linux

Cracking Files

- Protected Files
-

Protected Archives

Password Management

Password Policies

Password Managers

Skills Assessment

- Password Attacks Lab - Easy
- Password Attacks Lab - Medium
- Password Attacks Lab - Hard

My Workstation

OFFLINE

Start Instance

∞ / 1 spawns left